DO NOW — RECALL

Processor and memory synoptic challenge

Name: _____ Class: _____ Date: _____

1. What is an ADC used for?

2. Which register holds the current instruction?

3. What is the main limitation of two cores on wholly dependent steps?

4. A full adder receives 1,1 and carry-in 0. Give sum and carry-out and explain.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Converting an analogue sensor signal into a digital representation.

2. Model answer

CIR.

3. Model answer

The next step cannot start until the earlier result exists.

4. Model answer

$1+1+0=10_2$; sum bit 0, carry-out bit 1.

Vocabulary — know these terms

PC — Program Counter; next instruction address.

MAR — Memory Address Register.

MDR — Memory Data Register.

cache — Fast nearby memory.

throughput — Work completed per unit time.

NEW LEARNING

Read, annotate and explain

Fetch the instruction

PC identifies address 72, which MAR sends to memory. MDR transfers its contents to CIR.

Differentiate performance improvements

Cores can execute independent work; cache helps when needed data/instructions are present.

Explain limits

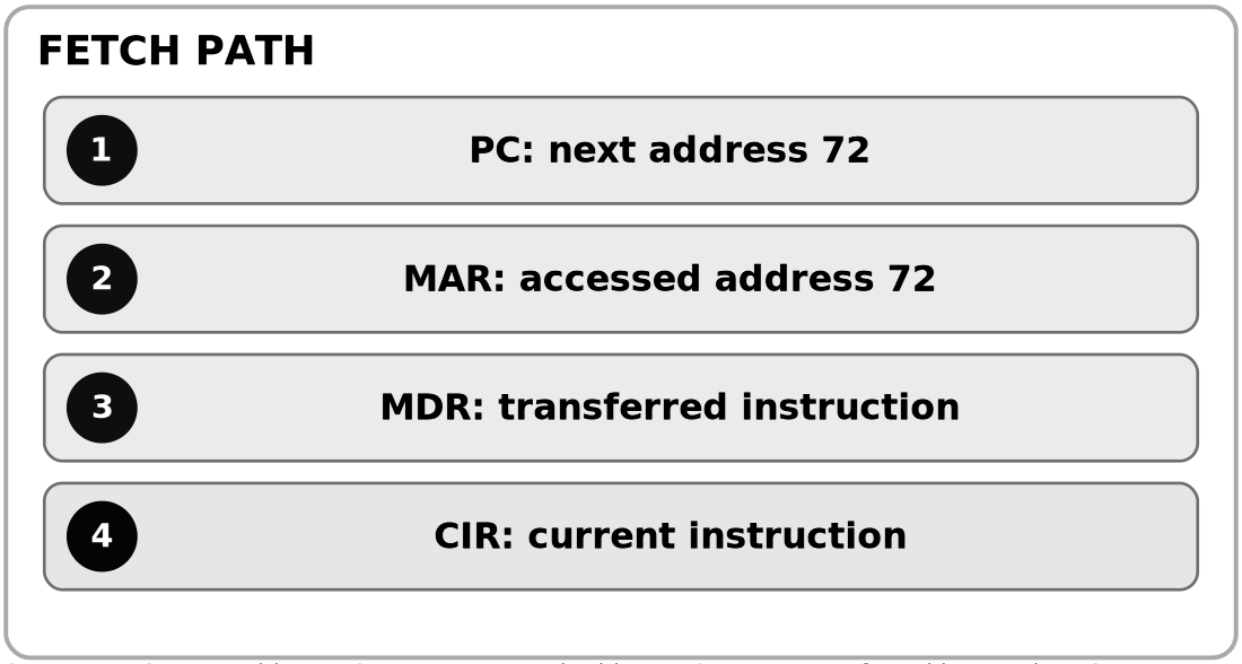
Branching, shared memory bandwidth, serial dependencies and software constrain real speedup.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

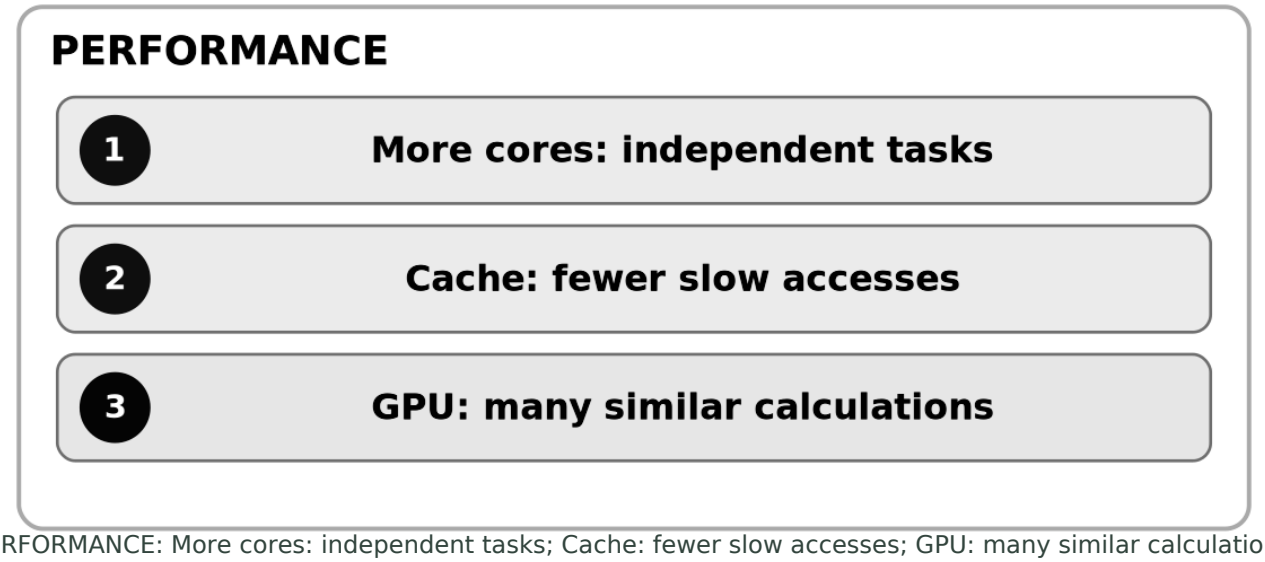
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FETCH PATH: PC: next address 72; MAR: accessed address 72; MDR: transferred instruction; CIR: current instruction

New Learning



PERFORMANCE: More cores: independent tasks; Cache: fewer slow accesses; GPU: many similar calculations

I DO — WORKED EXAMPLE

Follow the worked method

Starting PC=40, trace MAR, CIR and PC after a sequential fetch.

Predict

Copy PC=40 into MAR to select the instruction location.

Explain

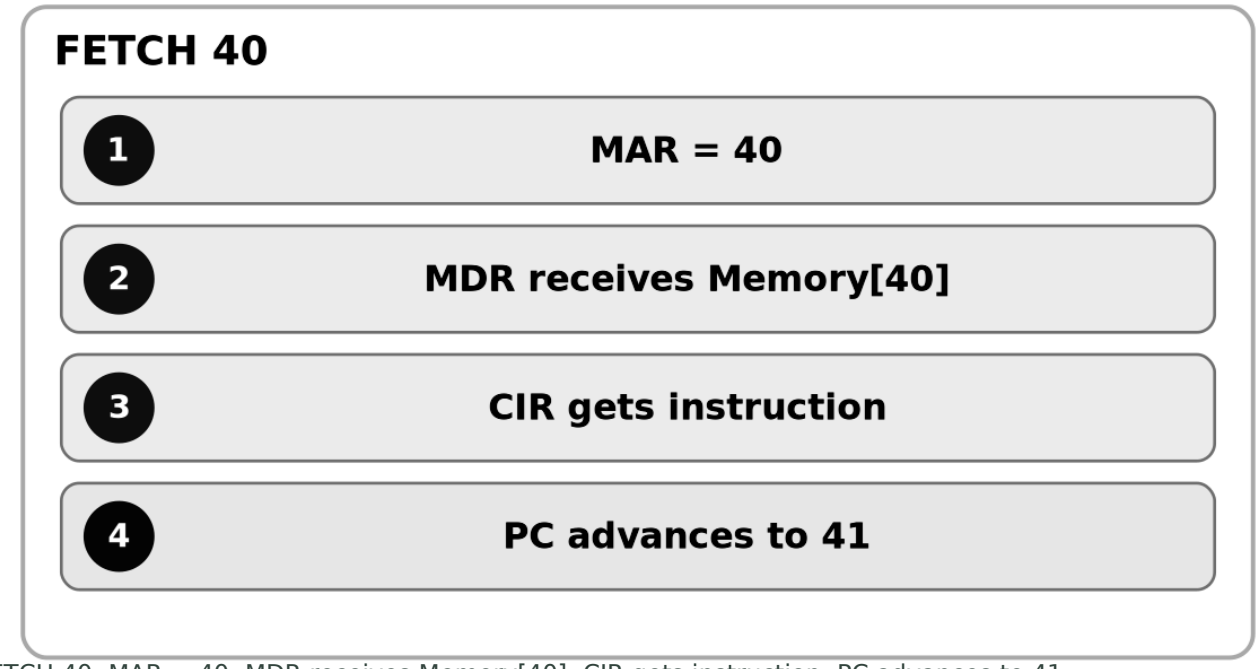
Instruction contents travel via MDR to CIR, while sequential PC advances to 41.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Starting PC=40, trace MAR, CIR and PC after a sequential fetch.

Copy PC=40 into MAR to select the instruction location.



FETCH 40: MAR = 40; MDR receives Memory[40]; CIR gets instruction; PC advances to 41

WE DO — GUIDED PRACTICE

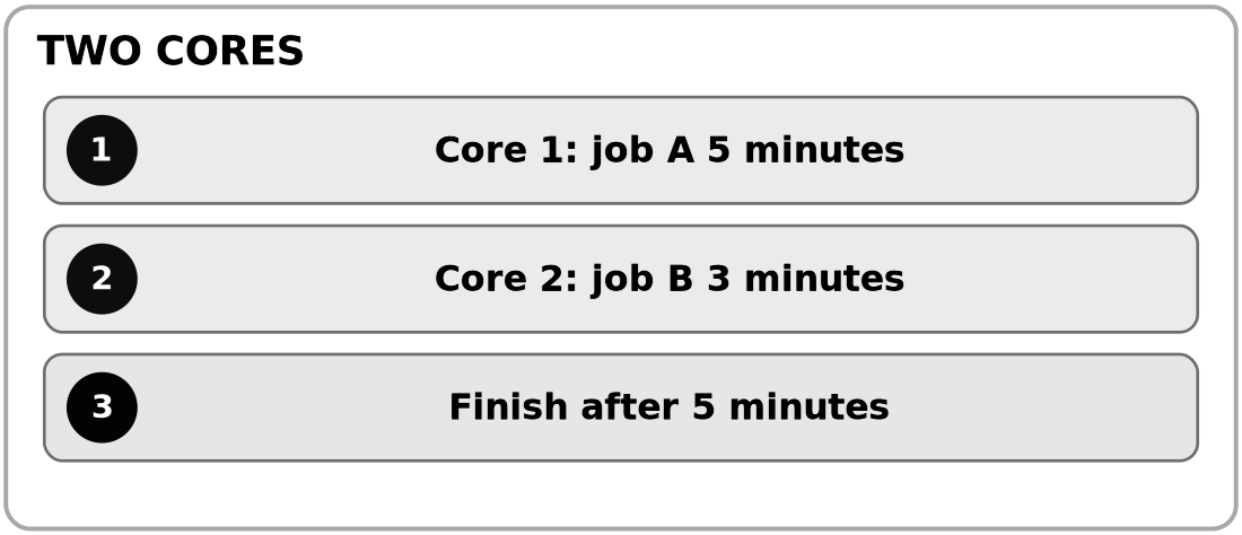
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two independent jobs take 5 and 3 minutes. Calculate ideal two-core completion time.

Each available core receives a separate independent job.



TWO CORES: Core 1: job A 5 minutes; Core 2: job B 3 minutes; Finish after 5 minutes

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Each available core receives a separate independent job.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State what PC, MAR, MDR, CIR and ACC contain or do.

Question 2

Explain how cache and extra cores could help video rendering, and why speedup may be less than expected.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare a multicore CPU and GPU for an interactive logic controller and a million independent pixels.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Doubling CPU cores always halves runtime.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

Ideal duration is the longer job, five minutes; serial execution would take eight.

IMPROVE: Explain one correction to your own reasoning.